Unit-1

Prepare a PPT/Poster about networking a lab with 90 systems and list out the issues identified in planning.

Networking a Lab with 90 Systems

Introduction

A Local Area Network (LAN) is a group of computers and other devices that are connected together within a limited area. LANs are used in schools, businesses, and other organizations to share resources such as printers, files, and internet access.

This poster will discuss the steps involved in networking a lab with 90 systems. We will also identify some of the issues that need to be considered when planning a LAN.

Steps in Networking a Lab with 90 Systems

1. Planning

   The first step is to plan the network. This includes determining the size of the network, the types of devices that will be connected, and the security requirements.

2. Wiring

   Once the plan is in place, the next step is to wire the network. This involves running cables between the computers and the network devices.

3. Installing the Network Devices

   The next step is to install the network devices. This includes routers, switches, and firewalls.

4. Configuring the Network Devices

   Once the network devices are installed, they need to be configured. This includes setting up IP addresses, subnet masks, and default gateways.

5. Testing the Network

Once the network is configured, it needs to be tested. This includes making sure that all of the devices are connected and that they can communicate with each other.

Issues to Consider

When planning a LAN, there are a number of issues that need to be considered. These include:

- The size of the network
- The types of devices that will be connected
- The security requirements
- The budget

Conclusion

Networking a lab with 90 systems can be a complex task. However, by following the steps outlined in this poster, you can successfully network your lab and provide your users with access to shared resources.

Issues Identified in Planning

- The size of the network: A network with 90 systems is a relatively large network. This means that you will need to use a high-bandwidth network to ensure that all of the devices can communicate with each other effectively.
- The types of devices that will be connected: The types of devices that will be connected to the network will also affect the design of the network. For example, if you will be connecting printers to the network, you will need to make sure that the network has enough bandwidth to support them.
- The security requirements: The security requirements of the network will also need to be considered. For example, if you will be connecting sensitive data to the network, you will need to implement security measures to protect that data.
- The budget: The budget will also be a factor in the design of the network. You will need to make sure that the network is affordable while still meeting the needs of your users.

Conclusion

By considering the issues outlined in this poster, you can successfully network your lab with 90 systems. This will provide your users with access to shared resources and improve the efficiency of your lab.